# Midterm Exam

**( 6:35 PM – 7:50 PM : 75 Minutes )**

- This exam will account for 20% of your overall grade.
- There are five (5) questions, worth 75 points in total. Please answer all of them.
- This is a *closed book*, *closed notes* exam. *No cheat sheets* are allowed.
- You are *not allowed to use your own calculator*. A scientific calculator will be available inside the Respondus Lockdown Browser.
- You are allowed to *use scratch papers* for your calculations.

## Good Luck!

| Question | Parts | Points |
|---|---|---|
| 1. Asymptotic Bounds | $(a), (b), (c)$ | $5 + 5 + 5 = 15$ |
| 2. Recurrences | $(a), (b), (c)$ | $5 + 5 + 5 = 15$ |
| 3. Longest Increasing Subsequence | – | 12 |
| 4. Selection Sort | – | 18 |
| 5. Integer Multiplication | $(a), (b)$ | $10 + 5 = 15$ |
| Total | | 75 |

**Question 1. [ 15 Points ] Asymptotic Bounds.** For each of the following statements state whether it is *True* or *False*. You must justify each answer. Assume that all log's are of base 2.

(*a*) [ **5 Points** ] $\sqrt{\log n} = \mathcal{O}\left(n^{\frac{\log \log n}{\log n}}\right)$

(*b*) [ **5 Points** ] $4^{\sin^2 n} = \Omega\left(\log^2 n\right)$

(*c*) [ **5 Points** ] $2^{2n^2} = \Theta\left(2^{3n^2}\right)$

**Question 2.** [ **15 Points** ] **Recurrences.** For each of the functions below either solve it using the Master Theorem or state why the Master Theorem does not apply.

(*a*) [ **5 Points** ] $T(n) = \begin{cases} \Theta(1), & \text{if } n \leq 1; \\ 2T\left(\frac{n}{2}\right) + \frac{1}{2}T(n) + n^2 \log^2 n, & \text{otherwise.} \end{cases}$

(*b*) [ **5 Points** ] $T(n) = \begin{cases} \Theta(1), & \text{if } n \leq 1; \\ 6T\left(\frac{n}{3}\right) + (3 + \sin n)\, n^3, & \text{otherwise.} \end{cases}$

(*c*) [ **5 Points** ] $T(n) = \begin{cases} \Theta(1), & \text{if } n \leq 1; \\ 2T\left(\frac{n}{2}\right) + 2T\left(\frac{n}{3}\right) + \Theta(n), & \text{otherwise.} \end{cases}$

**Question 3.** **[ 12 Points ] Longest Increasing Subsequence.** The LIS algorithm given below computes and returns the length of the *Longest Increasing Subsequence* in the input array $A[1..n]$ containing $n$ numbers.

```
LIS ( A[1..n], n )
allocate temporary array L[1..n]
L[1] = 1
for i = 2 to n do
   L[i] = 0
for i = 2 to n do
   for j = 1 to i − 1 do
      if (A[j] < A[i]) and (L[j] > L[i]) then
         L[i] = L[j]
   L[i] = L[i] + 1
maxL = L[1]
for i = 2 to n do
   if (L[i] > maxL) then
      maxL = L[i]
return maxL
```

Analyze the worst-case running time of LIS on an input of size $n$.

**Question 4.** [ **18 Points** ] **Selection Sort.** Given an array $A[1..n]$ of $n$ distinct numbers as input the *Selection Sort* algorithm rearranges the items of $A$ in nondecreasing order of value from left (smaller array index) to right (larger array index). Recall that the outermost ***for*** loop of the algorithm iterates from $j = 1$ to $j = A.length - 1$.

Suppose the following array $A[1..10]$ is given as input to the algorithm. Please show the state of this array after each iteration of the outermost ***for*** loop.

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **A** | 7 | 13 | 10 | 5 | 19 | 14 | 11 | 12 | 9 | 1 |

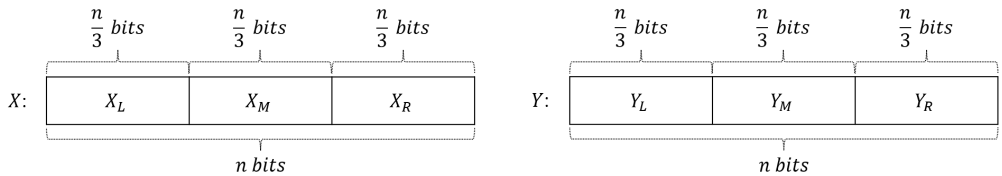
There must be one line for each value of $j$. Please start each line with the $j$ value followed by a ":" and then write down the items (separated by commas, no fancy boxes are needed) of the input array in the order they will appear in $A[1..10]$ at the end of the $j$-th iteration of the loop.

**QUESTION 5.** [ **15 Points** ] **Integer Multiplication.** Karatsuba's algorithm multiplies two $n$-bit integers by recursively computing pairwise products of $\frac{n}{2}$-bit integers. This question asks you to analyze the running time of an algorithm that solves the same problem by recursively multiplying $\frac{n}{3}$-bit integer pairs.

**PROBLEM:** Given two $n$-bit integers $X$ and $Y$, where $n = 3^k$ for integer $k \geq 0$, compute $Z = XY$.

**ALGORITHM:** If $n = 1$ then set $Z = XY$ and return. Otherwise, do the following.

Let $X = 2^{\frac{2n}{3}} X_L + 2^{\frac{n}{3}} X_M + X_R$ and $Y = 2^{\frac{2n}{3}} Y_L + 2^{\frac{n}{3}} Y_M + Y_R$.




**Step 1:**

$$\begin{aligned}
P_0 &= X_L Y_L \\
P_1 &= X_R Y_R \\
P_2 &= (X_L + X_M + X_R)(Y_L + Y_M + Y_R) \\
P_3 &= (X_L - X_M + X_R)(Y_L - Y_M + Y_R) \\
P_4 &= (4X_L - 2X_M + X_R)(4Y_L - 2Y_M + Y_R)
\end{aligned}$$

**Step 2:**

$$\begin{aligned}
Z_4 &= P_0 \\
Z_3 &= \tfrac{P_3 - P_1}{2} + \tfrac{P_2 - P_4}{6} + 2P_0 \\
Z_2 &= \tfrac{P_2 + P_3}{2} - P_0 - P_1 \\
Z_1 &= \tfrac{P_1 + P_2}{2} - \tfrac{P_2 - P_4}{6} - 2P_0 - P_3 \\
Z_0 &= P_1
\end{aligned}$$

**Step 3:**

$$Z = 2^{\frac{4n}{3}} Z_4 + 2^n Z_3 + 2^{\frac{2n}{3}} Z_2 + 2^{\frac{n}{3}} Z_1 + Z_0.$$

Now answer the following questions.

($a$) [ **10 Points** ] Write a recurrence relation describing the running time of the algorithm above when multiplying two $n$-bit integers. You can assume that each of $\frac{P_3 - P_1}{2}$, $\frac{P_2 + P_3}{2}$, $\frac{P_1 + P_2}{2}$, and $\frac{P_2 - P_4}{6}$ can be computed in $\Theta(n)$ time.

($b$) [ **5 Points** ] Solve the recurrence relation from part ($a$).